

Assignment 13: Natural Language Processing with Python

Name Ayleen Santander

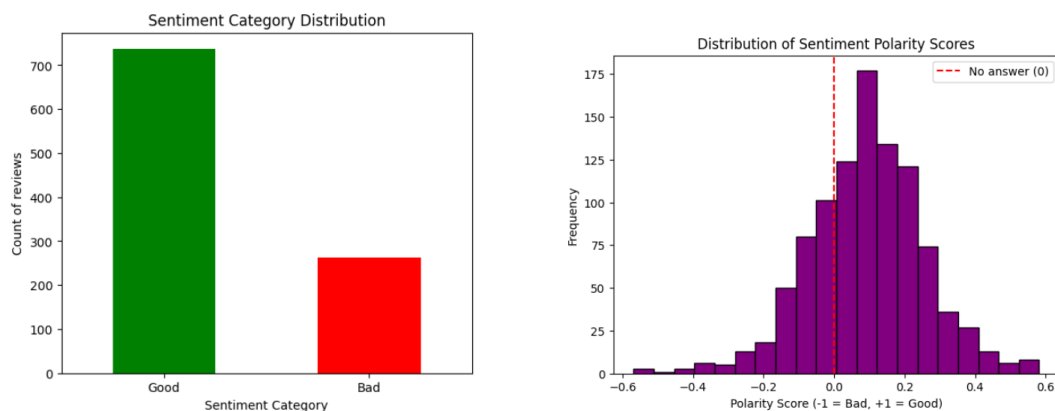
Part 2: Sentiment Analysis Task

1. Sentiment Distribution and Insights

The full dataset contains 50,000 reviews with extensive text content, labeled by sentiment: Positive, Neutral, and Negative. For this analysis, a random sample of 1,000 rows was selected as a representative subset. A cloned copy of the data was used with the sentiment label held out, so the exploratory analysis would not be biased by knowing the "correct" answer in advance.

The distribution across the 1,000-row sample is:

- **Good (Positive): 734 reviews (73.4%)**
- **Bad (Negative): 264 reviews (26.4%)**
- **No answer (unlabeled/Neutral): 2 reviews (0.2%)**



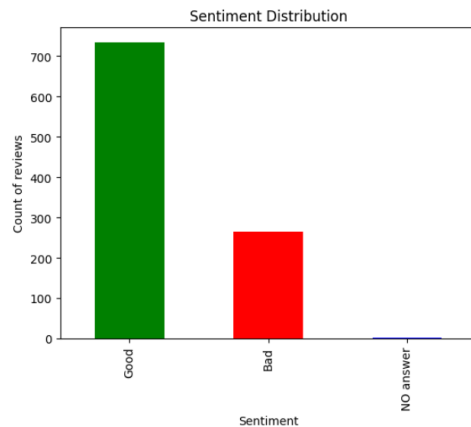
This is a clearly imbalanced dataset, roughly 3 out of every 4 reviews are positive. The "Sentiment Category Distribution" chart visualizes this gap between Good and Bad directly, while the "Distribution of Sentiment Polarity Scores" chart confirms it numerically: the polarity scores skew right (toward positive), with the bulk of the distribution sitting above the neutral midline. The "Sentiment Distribution" chart shows all three categories together, including the negligible "No answer" group.

While positive reviews dominate, the 264 negative reviews represent a meaningful signal worth investigating further, understanding *why* customers leave negative reviews is more actionable for the business than confirming that most customers are satisfied. That question (what drives negative sentiment) is the direction for deeper analysis beyond this initial distribution check.

```

sentiment_category
Good      734
Bad       264
NO answer    2
Name: count, dtype: int64

```



2. Model Training: Naive Bayes vs. SVM

Reviews labeled "No answer" were excluded from classification, since two examples are too few to train or evaluate a model reliably. The remaining Good/Bad labeled reviews were split into training and test sets (200 held out for testing: 147 Good, 53 Bad), and two classifiers were trained: Naive Bayes and Support Vector Machine (SVM).

```

Naive Bayes Classifier:
precision    recall  f1-score   support

   Bad         0.00    0.00    0.00         53
   Good         0.73    1.00    0.85        147

 accuracy          0.73         200
 macro avg         0.37    0.50    0.42         200
 weighted avg         0.54    0.73    0.62         200

Accuracy: 0.735
-----SVM-----
precision    recall  f1-score   support

   Bad         0.78    0.40    0.53         53
   Good         0.82    0.96    0.88        147

 accuracy          0.81         200
 macro avg         0.80    0.68    0.70         200
 weighted avg         0.81    0.81    0.79         200

Accuracy: 0.81

```

Naive Bayes: Accuracy: 73.5%

This looks reasonable at first glance, but the underlying behavior tells a different story. Naive Bayes achieved 0.00 precision and 0.00 recall on the "Bad" class, meaning it never once predicted a review as negative. It effectively defaulted to guessing "Good" for every single review. Since 73.5% of the test set is Good, this strategy alone produces 73.5% accuracy without the model learning anything. The high accuracy number is misleading; the model has no ability to detect negative sentiment at all.

SVM: Accuracy: 81%

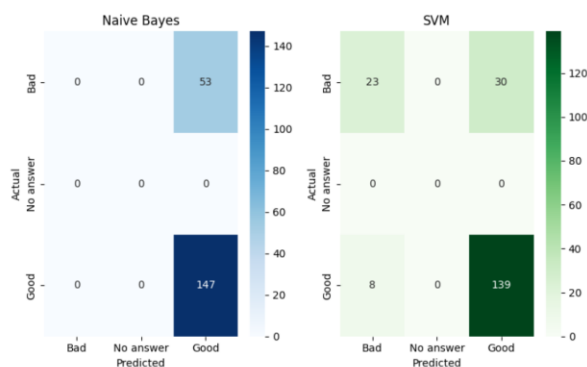
SVM performs meaningfully better and, more importantly, learns to distinguish the two classes. On the "Bad" class, it reaches 78% precision (when it predicts "Bad," it's usually right) and 40%

recall (it catches 40% of the true negative reviews). On "Good," precision is 82% and recall is 96%. The macro-averaged F1-score (which weights both classes equally, ignoring their size) is 0.70 for SVM versus just 0.42 for Naive Bayes, this is the clearest evidence that SVM has real discriminative power while Naive Bayes does not.

3. Confusion Matrix and Business Interpretation

The confusion matrix makes this gap visible directly: Naive Bayes misclassifies every single Bad review as Good, while SVM correctly identifies a meaningful portion of them, though it still misses more than half (60 point recall gap on Bad).

This has a direct business implication: a negative review that gets misclassified as positive is a missed warning sign. In commercial terms, **failing to catch negative feedback carries more risk than failing to catch positive feedback**, a company that can't detect dissatisfied customers can't act on that feedback, which affects retention and future sales. This asymmetry is why accuracy alone is not sufficient metric here; recall on the "Bad" class matters more than overall accuracy for this business problem.



4. Model Comparison Against Actual Reviews

Comparing both models' predictions to the actual labeled reviews confirmed that SVM aligns with ground truth more often. SVM misclassified 38 out of 200 test reviews, compared to Naive Bayes' 53 misclassifications (all of them Bad reviews predicted as Good). When the two models' predictions were compared directly against each other, they disagreed on 27 reviews, a useful check that highlights where the models' decision boundaries diverge, and a natural starting point for error analysis.

review actual predicted_nb predicted_svm					38 misclassified out of 200	review actual predicted_svm		
0	ye lous film purple butterfly pit secret organ...	Bad	Good	Good	0	ye lous film purple butterfly pit secret organ...	Bad	Good
1	good lord think spoiler warn even though think...	Bad	Good	Bad	4	comment first debut give thumb middle review r...	Bad	Good
2	film mature educate audience saw random heart ...	Good	Good	Good	7	like go see film history class something like ...	Bad	Good
3	karen carpenter story show little singer karen...	Good	Good	Good	11	think mukhsin wonderfully write entertainment ...	Bad	Good
4	comment first debut give thumb middle review r...	Bad	Good	Good	13	great ridiculous horror movie capture essence ...	Bad	Good
5	francis ford coppola write direct stunningly p...	Good	Good	Good	22	one bad movie ever see comedy funny also badly...	Bad	Good
6	okay watched movie twice research heavily net ...	Good	Good	Good	24	lorne michael film seem fail essentially exten...	Bad	Good
7	like go see film history class something like ...	Bad	Good	Good	25	write oliver stone direct brian de palma scarf...	Bad	Good
8	taut organically grip edward dmytryk's crossfir...	Good	Good	Good	30	first realize rating suppose reserve bad bad m...	Good	Bad
9	pepe le moko play charles boyer sort internati...	Good	Good	Good	34	hollywood movie industry lazy one entire world...	Bad	Good

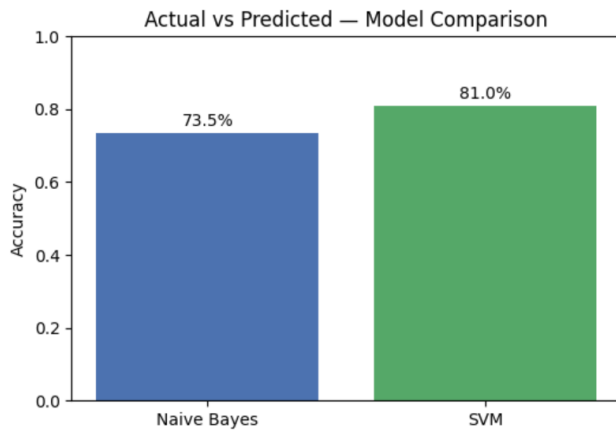
And when compared models disagree with each other the output on 31 reviews are disagree between them.

Models disagree on 31 reviews

	review	actual	predicted_nb	predicted_svm	
1	open shot yous marine seriously disrespect you...	Bad	Good	Bad	
7	get wrong love bad movie adore bad movie troll...	Bad	Good	Bad	
13	sorry say film indeed bad reminds cgrade porn ...	Bad	Good	Bad	
30	first realize rating suppose reserve bad bad m...	Good	Good	Bad	
31	abc gear repertory company another unrealistic...	Bad	Good	Bad	
35	besides boring scene oppressive dark movie try...	Good	Good	Bad	
43	mighty morphin power ranger get bad television...	Bad	Good	Bad	
63	scotty grant cramer would go star great bmovie...	Good	Good	Bad	
67	watch first episodes short back felt could tak...	Bad	Good	Bad	
72	movie come huge disappointment anime series en...	Bad	Good	Bad	

5. Conclusion part 1

Based on overall accuracy alone, SVM is the stronger model (81% vs. 73.5%). But the more important finding is that Naive Bayes' accuracy is an illusion created by class imbalance, it fails completely at the actual task of detecting negative sentiment. SVM, while imperfect (missing 60% of true negative reviews), is the only model that demonstrates real learning on both classes, supported by its substantially higher macro F1-score (0.70 vs. 0.42).



Preliminary Recommendation: Given that negative reviews are the higher-value signal for the business, next steps should focus on improving recall on the "Bad" class specifically, for example, using `class_weight='balanced'` in SVM, applying oversampling techniques (e.g., SMOTE) to the minority class, or running topic modeling on the 263 negative reviews to identify recurring complaint themes that could inform product or service improvements.

3. Model VADER and Analysis: (applied the model after)

The model took the original reviews, cleaned punctuation and capitalization, and generated a `sentiment_score` for each. Using this score, reviews were classified into Bad, No Answer, and Good, and we could determine how each category is represented as a percentage of the total.

Totaling the sentiment categories under VADER:

- **Good:** 625 (62.5%)
- **Bad:** 371 (37.1%)
- **No Answer:** 4 (0.4%)

	review	sentiment_score
0	one of the other reviewers has mentioned that ...	{'neg': 0.195, 'neu': 0.749, 'pos': 0.056, 'co...
1	a wonderful little production. the...	{'neg': 0.053, 'neu': 0.776, 'pos': 0.172, 'co...
2	i thought this was a wonderful way to spend ti...	{'neg': 0.094, 'neu': 0.714, 'pos': 0.192, 'co...
3	basically there's a family where a little boy ...	{'neg': 0.138, 'neu': 0.797, 'pos': 0.065, 'co...
4	petter mattei's "love in the time of money" is...	{'neg': 0.052, 'neu': 0.801, 'pos': 0.147, 'co...
...	...	...
995	i really love this movie. i remember one time ...	{'neg': 0.101, 'neu': 0.811, 'pos': 0.088, 'co...
996	upon the straight story release in 1999, it wa...	{'neg': 0.108, 'neu': 0.737, 'pos': 0.155, 'co...
997	as a massive fan of the three tv series, i was...	{'neg': 0.107, 'neu': 0.733, 'pos': 0.16, 'com...
998	in the veins of jeepers creepers and the texas...	{'neg': 0.181, 'neu': 0.644, 'pos': 0.174, 'co...
999	now don't get me wrong i love bad movies... no...	{'neg': 0.265, 'neu': 0.554, 'pos': 0.181, 'co...

1000 rows × 2 columns

When we totalize the sentiment category the output shows: **Good = 625 ~ 62.5%, Bab= 371 ~ 37.1% and No Answer = 4 ~ 0.4%**, those numbers are very different with the previous sentiment methods

```

• sentiment_category
  good          625
  bad           371
  no answer      4
  Name: count, dtype: int64

```

These numbers differ noticeably from the original labeled distribution (73.4% / 26.4% / 0.2%), VADER classifies a meaningfully larger share of reviews as negative.

```

sentiment_category
Good          734
Bad           264
NO answer      2
Name: count, dtype: int64

```

Using the compound score, results were cleaned and standardized, then converted into sentiment labels, this makes the underlying distribution more visible than the raw category counts. The most extreme scores in the dataset were -0.9947 (extremely negative) and 0.9641

(extremely positive); reviews scoring near the middle were labeled "neutral" (No Answer).

	review	bad	no answer	good	compound	
0	one of the other reviewers has mentioned that ...	0.195	0.749	0.056	-0.9947	Sentiment analysis results:
1	a wonderful little production. the...	0.053	0.776	0.172	0.9641	review good bad compound \
2	i thought this was a wonderful way to spend ti...	0.094	0.714	0.192	0.9605	0 one of the other reviewers has mentioned that ... 0.056 0.195 -0.9947
3	basically there's a family where a little boy ...	0.138	0.797	0.065	-0.9213	1 a wonderful little production. the... 0.172 0.053 0.9641
4	petter matter's "love in the time of money" is...	0.052	0.801	0.147	0.9744	2 i thought this was a wonderful way to spend ti... 0.192 0.094 0.9605
...	...	...	...	...	...	3 basically there's a family where a little boy ... 0.065 0.138 -0.9213
995	i really love this movie. i remember one time ...	0.101	0.811	0.088	-0.0872	4 petter mattei's "love in the time of money" is... 0.147 0.052 0.9744
996	upon the straight story release in 1999, it wa...	0.108	0.737	0.155	0.9808	sentiment_category
997	as a massive fan of the three tv series, i was...	0.107	0.733	0.160	0.9325	0 bad
998	in the veins of jeepers creepers and the texas...	0.181	0.644	0.174	-0.5423	1 good
999	now don't get me wrong i love bad movies... no...	0.265	0.554	0.181	-0.9867	2 good
...	...	...	...	...	...	3 bad
...	...	...	...	...	...	4 good

No we need to identify the percentage of each reviews, below it shows the evaluation results:

=====	
Sentiment Evaluation:	
=====	
Sentiment Counts:	
sentiment_category	
good 625	
bad 371	
no answer 4	
Name: count, dtype: int64	
Sentiment Percentages:	
sentiment_category	
good 62.5	Sentiment Distribution
bad 37.1	good: 625 records(62.50%)
no answer 0.4	bad: 371 records(37.10%)
Name: proportion, dtype: float64	no answer: 4 records(0.40%)

The overall average compound score across all reviews is 0.24, putting the aggregate sentiment as Good; for example, if this were a product or customer-service survey, the overall takeaway would be that feedback skews positive. In this case, since the data is movie reviews, that translates to overall favorable reception.

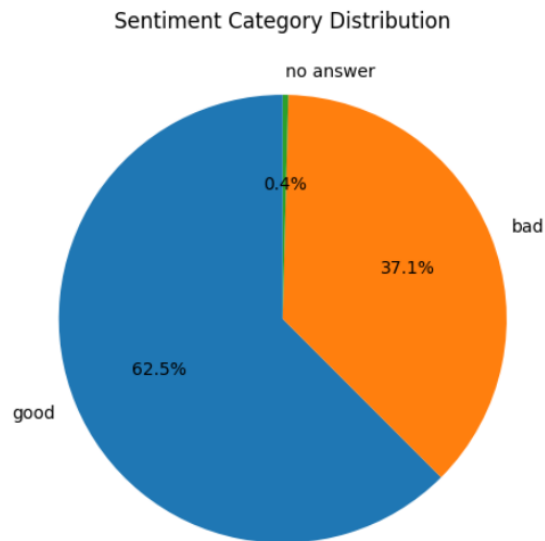
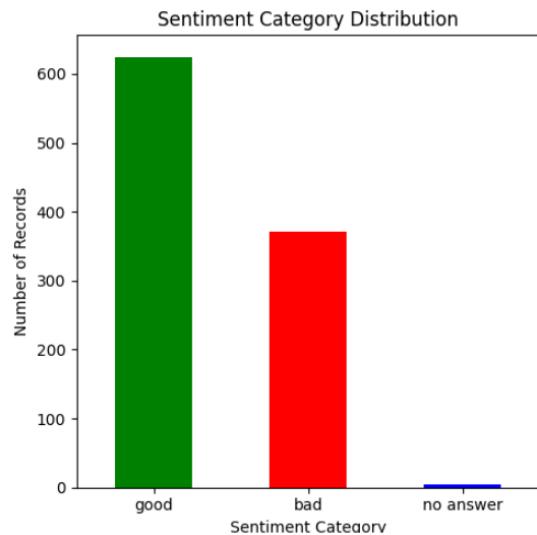
Average Compound Score: 0.24
Overall Sentiment: Good

Even though most reviews are positive, the negative reviews remain the more useful group to study, since they're what would point to what a business needs to improve, or where a team needs more training or attention.

```
Most Good Review:
review      i firmly believe that the best oscar ceremony ...
compound                                         0.9996
Name: 410, dtype: object
Score: 0.9996
```

```
Most Bad Review:
review      oh, my goodness. i would have never thought it...
compound                                         -0.9994
Name: 456, dtype: object
Score: -0.9994
```

Visualization makes the gap between Good and Bad reviews clearly visible.



4. Applying Naive Bayes and SVM with VADER Labels

When both models were retrained using the VADER-derived labels, Naive Bayes' accuracy dropped to 63.5%, lower than SVM's 75.5%, both lower than in Part 2, and neither high enough to fully trust.

Naive Bayes Analysis:

- **Precision:** Bad = 1.00 (every review it predicted as Bad actually was Bad); Good = 0.63 (moderate confidence); No Answer = 0.0 (never correctly predicted).
- **Recall:** Bad = 0.03, this is a **poor** result: the model only caught 3% of the actual Bad reviews, meaning it almost never predicts "Bad" at all. Good = 1.00 (it catches essentially all true Good reviews); No Answer = 0.0.
- **F1-Score:** Consistently low across classes, reflecting the imbalance between precision and recall above.
- **Support:** Bad = 74, Good = 125, No Answer = 1.

This is a notably different failure mode below. There, Naive Bayes never predicted "Bad" at all (0.00 precision and recall). Here, it rarely predicts "Bad," but is highly accurate on the few times it does (precision 1.00), while still missing almost every true negative review (recall 0.03). Either way, the model is not usable for catching negative sentiment.

SVM Analysis:

- **Precision:** Bad = 0.72, Good = 0.77, No Answer = 0.0 (too few examples to learn from).
- **Recall:** Bad = 0.58, (still missing a large share of true negative; Good = 0.86 (catching most, but not all, positives)).
- **F1-Score:** Still lower than in the previous analysis, reflecting the noisier VADER-based labels.
- **Support:** Bad = 74, Good = 125, No Answer = 1.

Naive Bayes Classifier:				
	precision	recall	f1-score	support
bad	1.00	0.03	0.05	74
good	0.63	1.00	0.77	125
no answer	0.00	0.00	0.00	1
accuracy			0.64	200
macro avg	0.54	0.34	0.28	200
weighted avg	0.76	0.64	0.50	200

Accuracy: 0.635

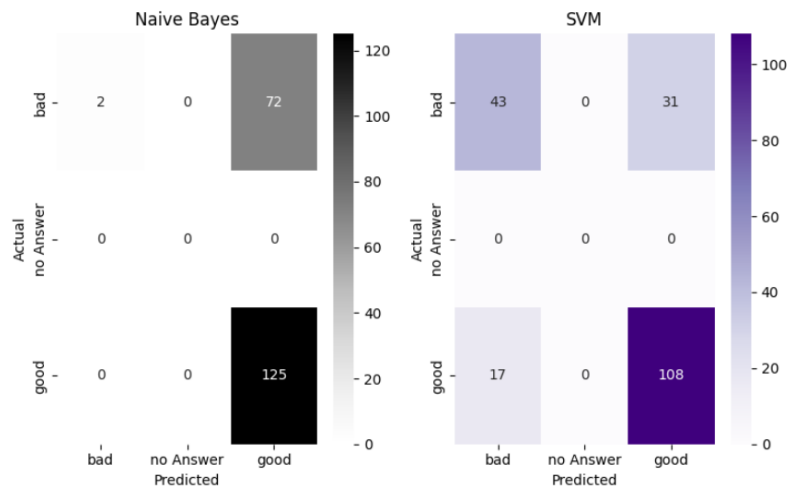
-----SVM-----

	precision	recall	f1-score	support
bad	0.72	0.58	0.64	74
good	0.77	0.86	0.82	125
no answer	0.00	0.00	0.00	1
accuracy			0.76	200
macro avg	0.50	0.48	0.49	200
weighted avg	0.75	0.76	0.75	200

Accuracy: 0.755

When comparing SVM's predictions to the actual VADER labels, SVM misclassified 49 out of 200 reviews. When comparing Naïve Bayes and SVM's predictions to each other, the two models disagreed on 58 reviews.

Business Pattern: Across both the original labels (previous part) and VADER-derived labels (part 3), the same pattern holds, models struggle more to catch negative reviews than positive ones, and negative reviews carry disproportionate business weight: a handful of missed negative reviews has more impact on customer retention and future sales than an equivalent number of missed positive reviews.



Naive Bayes accuracy: 0.635
SVM accuracy: 0.755

When we tried to identify the misclassification in the reviews with SVM classifiers contain 49 misclassified out of 200.

49 misclassified out of 200

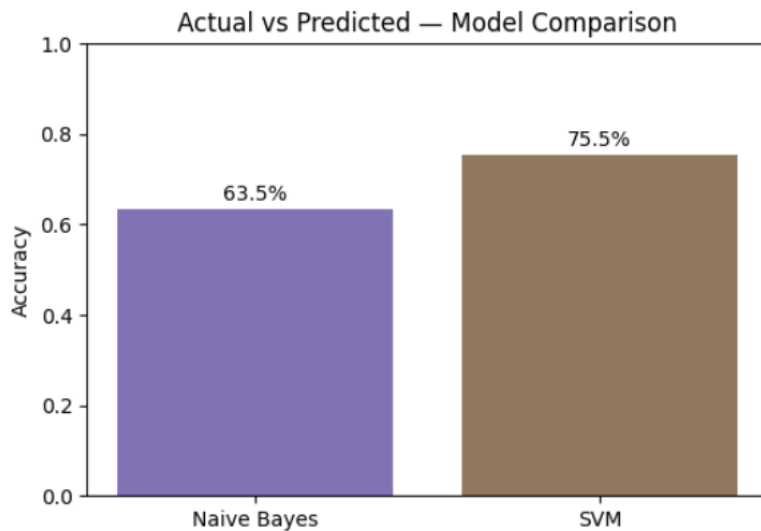
	review	actual	predicted_svm
2	like armand assante cable company summary soun...	bad	good
5	may go film like assuredly college professor d...	bad	good
13	saw movie last year medium class say really ha...	bad	good
15	open shot yours marine seriously disrespect you...	bad	good
16	deny affinity speed movie like fast furious dh...	bad	good
32	say event hit saw documentary take year come g...	bad	good
37	percent nation nitwit still support w bush wou...	bad	good
39	difficult tough brad sykes hardworking guy lov...	good	bad
44	incomprehensibly dreadful mishmash probably no...	bad	good
49	filmmaker want make movie without story tell r...	bad	good

Comparison with both models disagree with each other, model disagree on 58 reviews, this means Naïve Bayes indicated 58 were misclassified.

- Models disagree on 58 reviews

	review	actual	predicted_nb	predicted_svm
4	sometimes patient watch indie horror toetap wa...	bad	good	bad
7	mockumentaries proliferate lately much approac...	bad	good	bad
10	number video nasties yet see dwindles little p...	bad	good	bad
12	please rewatch series go see movie trailer com...	bad	good	bad
17	get back free screening osama witch project ha...	bad	good	bad
21	first reason give film star instead least pete...	bad	good	bad
27	one say warn read review user external like u ...	bad	good	bad
29	ever wonder bad movie ever make stop wondering...	bad	good	bad
34	buy film blockbuster sound interesting bit ran...	bad	good	bad
38	basically family little boy jake think zombie ...	bad	good	bad

Final comparison between Naïve Bayes and SVM models



Conclusion and Recommendations:

Baded on both analyses, SVM consistently outperforms Naïve Bayes at detecting negative sentiment, though neither model reaches strong recall on the 'Bad' class, and VADER-derived labels produced lower accuracy for both models than the original ground-truth labels, likely because VADER's scoring is noisier than human-assigned labels. The consistent finding across both labels sets is that class imbalance and weak 'Bad' class recall are the central weaknesses to address, through techniques like class weight = 'balanced', SMOTE oversampling, or a larger negative-review sample size.